

Definition of tokenization:

Tokenization is the process of exchanging sensitive data for nonsensitive data called “tokens” that can be used in a database or internal system without bringing it into scope.

The original sensitive data is then safely stored outside of the organization’s internal systems.

Tokenization works by removing the valuable data from your environment and replacing it with these tokens. Most businesses hold at least some sensitive data within their systems, whether it be credit card data, medical information, Social Security numbers, or anything else that requires security and protection.

Using tokenization, organizations can continue to use this data for business purposes without incurring the risk or compliance scope of storing sensitive data internally.

Tokenization, which seeks to minimize the amount of sensitive data a business needs to keep on hand, has become a popular way for small and midsize businesses to bolster the security of credit card and [e-commerce](#) transactions while minimizing the cost and complexity of [compliance](#) with industry standards and government regulations.

function tokenization:

1. **Identification of Asset:** The first step in tokenization is identifying the asset to be tokenized. This could be any asset with value, such as real estate, stocks, commodities, artwork, or intellectual property rights.
2. **Legal and Regulatory Compliance:** Before tokenizing an asset, it's essential to ensure compliance with relevant laws, regulations, and securities requirements. Legal agreements and documentation may need to be drafted to establish ownership rights, transferability, and other terms and conditions.
3. **Creation of Digital Tokens:** Once the asset is identified and legal considerations are addressed, digital tokens representing ownership or rights to the asset are created on a blockchain platform. This typically involves the deployment of a smart contract that defines the token's characteristics, such as supply, divisibility, transferability, and other attributes.
4. **Tokenization Process:** The asset is then tokenized by associating each unit of ownership or rights with a corresponding digital token. These tokens are usually issued on a blockchain network, such as Ethereum, using standards like ERC-20 (for fungible tokens) or ERC-721 (for non-fungible tokens).
5. **Transfer and Trading:** Once tokenized, the digital tokens can be transferred, traded, or exchanged on blockchain-based platforms, such as decentralized exchanges (DEXs) or token issuance platforms. This enables fractional ownership, increased liquidity, and broader market access for the tokenized asset.

How is a Token Created in the Tokenization Process?

When a business partners with a third-party tokenization provider, the following 4 step process is initiated:

1. Data Transformation:

Sensitive data is exported and sent to the third-party tokenization provider, which is then transformed into nonsensitive placeholders called tokens. This process offers advantages over encryption, as tokenization does not rely on keys to modify the original data.

2. Extraction from Internal Systems:

Sensitive data is then extracted entirely from an organization's internal systems and replaced with tokens. This ensures the original sensitive information is no longer present internally.

3. Operational Use of Tokens:

The tokens, which are mathematically unrelated to the original data, are retained within the organization for operational purposes. This allows businesses to continue using the tokens for various activities without the risk of exposing sensitive information.

4. Secure External Storage:

The original sensitive data, which has been replaced by tokens internally, is securely stored outside the organization's environment by the third-party tokenization provider.

Advantages of Tokenization:

1. **Increased Liquidity:** Tokenization converts traditionally illiquid assets, such as real estate or fine art, into digital tokens that can be traded on blockchain-based platforms. This enhances liquidity by allowing fractional ownership and enabling assets to be bought and sold more easily.
2. **Accessibility:** Tokenization democratizes access to investment opportunities by lowering barriers to entry. Fractional ownership allows investors to participate in asset markets with smaller amounts of capital, opening up investment opportunities to a wider range of individuals.
3. **24/7 Market:** Digital tokens can be traded 24/7 on blockchain-based platforms, allowing for continuous market access and real-time trading. This eliminates the need for traditional market hours and enables investors to react to market conditions at any time.
4. **Efficiency:** Blockchain-based tokenization streamlines the process of asset issuance, transfer, and settlement. Smart contracts automate transaction execution, reducing the need for intermediaries, paperwork, and manual processes, thereby improving efficiency and reducing transaction costs.
5. **Fractional Ownership:** Tokenization allows assets to be divided into smaller units, enabling fractional ownership. This makes high-value assets more accessible to investors who may not have the capital to purchase them outright, leading to greater diversification and risk mitigation.
6. **Transparency and Security:** Tokenization on blockchain networks provides transparency, immutability, and security. Transaction records are recorded on the blockchain, providing a verifiable and tamper-resistant record of ownership and transaction history.

Disadvantages of Tokenization:

1. **Regulatory Complexity:** Tokenization of assets may be subject to complex regulatory requirements, depending on the jurisdiction and the nature of the asset. Compliance with securities regulations, anti-money laundering (AML) laws, and know-your-customer (KYC) requirements can pose challenges for tokenization projects.
2. **Lack of Standardization:** The tokenization space lacks standardized protocols and frameworks, leading to fragmentation and interoperability issues. Different platforms may use incompatible token standards or governance models, hindering the seamless transfer and trading of tokenized assets across platforms.
3. **Market Volatility:** Tokenized assets may be subject to market volatility, just like traditional financial markets. Price fluctuations can impact the value of tokenized assets and lead to potential losses for investors, especially in highly speculative markets or during periods of market instability.
4. **Cybersecurity Risks:** Blockchain-based tokenization introduces cybersecurity risks, including smart contract vulnerabilities, hacking attacks, and theft of digital assets. Security breaches or vulnerabilities in tokenization platforms can result in financial losses and undermine investor confidence in the tokenized assets.
5. **Custodial Risks:** Tokenization platforms may require users to entrust their digital assets to custodians or third-party service providers. Custodial arrangements introduce counterparty risks and reliance on trusted intermediaries, which may compromise the decentralization and security benefits of blockchain technology.

↓ Examples of tokenization:

Tokenization technology can, in theory, be used with sensitive data of all kinds, including bank transactions, medical records, criminal records, vehicle driver information, loan applications, stock trading and voter registration. For the most part, any system in which surrogate, nonsensitive information can act as a stand-in for sensitive information can benefit from tokenization.

Tokenization is often used to protect credit card data, bank account information and other sensitive data handled by payment processors. Payment processing use cases that tokenize sensitive credit card information include the following:

- mobile wallets, such as Google Pay and Apple Pay;
- e-commerce sites; and
- businesses that keep customers' cards on file

Token offerings:

ICO (Initial Coin Offering):

ICO is an event that helps generate funds for businesses and projects in the crypto or blockchain ecosystem by offering new digital tokens or coins to investors in exchange for other crypto or fiat cash.

The purpose of an ICO is to offer a mechanism for a business or project to get finance without using conventional financing channels such as venture capital or banks.

ICO coins constitute a more decentralized and democratic method of raising cash, enabling participation from anybody with an internet connection, regardless of geography or financial standing.

Initial coin offerings (ICOs) are a popular way to raise funds for products and services usually related to cryptocurrency.

ICOs are similar to initial public offerings (IPOs), but coins issued in an ICO also can have utility for a software service or product.

A few ICOs have yielded returns for investors. Numerous others have turned out to be fraudulent or have performed poorly.

To participate in an ICO, you usually need to first purchase a more established digital currency, plus have a basic understanding of cryptocurrency wallets and exchanges.

ICOs are, for the most part, completely unregulated, so investors must exercise a high degree of caution and diligence when researching and investing in them.

Examples of Initial Coin Offerings

Ethereum's ICO in 2014 is an early, prominent example of an initial coin offering. The Ethereum ICO raised \$18 million over a period of 42 days.¹¹ In 2015, a two-phase ICO began for a company called Antshares, which later rebranded as Neo. The first phase of this ICO ended in October 2015, and the second continued until September 2016. During this time, Neo generated about \$4.5 million.¹²

What is a Security Token Offering?

Security Token Offering (STO) is a fundraising mechanism that has gained significant popularity in the financial landscape, especially in the realm of blockchain-based projects.

STOs involve the issuance and sale of security tokens, which are digital assets that represent ownership or investment interests in a tangible asset or enterprise.

In contrast to Initial Coin Offerings (ICOs), which often involve utility tokens and lack regulatory oversight, STOs are subject to securities laws and regulations.

They offer a more secure and compliant way for businesses to raise capital by tokenizing assets and providing investors with tangible ownership rights.

Security tokens derive their value from real-world assets like equity in a company, real estate, fine art, or commodities. The value of these tokens is directly tied to the performance of the asset they represent, which can make them more stable and reliable compared to cryptocurrencies.

How STOs Work

Security Token Offerings follow a well-defined process that involves various stages:

Asset Tokenization: The STO process begins with the tokenization of real-world assets. These assets can be anything of value, such as real estate properties, company shares, precious metals, or even works of art. The assets' ownership rights or investment interests are then converted into digital securities represented by security tokens on a blockchain network.

Legal Compliance: Unlike Initial Coin Offerings (ICOs), which often operate in regulatory gray areas, STOs must comply with securities laws and regulations in the jurisdictions where they are conducted. This adherence to regulatory standards ensures investor protection and transparency throughout the offering process.

Investment Rounds: STOs are typically conducted in rounds or stages. Each round involves the issuance and sale of security tokens to investors in exchange for fiat currency or cryptocurrencies like Bitcoin or Ethereum. These tokens represent ownership stakes or financial interests in the underlying assets.

Smart Contracts: Blockchain technology plays a pivotal role in STOs through the use of smart contracts. These self-executing contracts automatically enforce the terms and conditions of the token sale and investment process. Smart contracts are programmed to govern factors such as investor rights, profit-sharing mechanisms, and the distribution of dividends.

Compliance Verification: During the STO process, participants, including investors and issuers, must undergo compliance verification. This often includes Know Your Customer (KYC) and Anti-Money Laundering (AML) checks to ensure all parties involved meet the necessary legal requirements.

Benefits of Security Token Offerings:

Enhanced Liquidity: One of the most significant benefits of STOs is the enhanced liquidity they offer. Traditional investment options, such as real estate or private equity, often suffer from illiquidity, making it challenging for investors to exit their positions quickly.

Access to Global Investor Base: By leveraging blockchain technology and conducting STOs, businesses gain access to a global pool of investors. Traditional fundraising methods typically involve geographic limitations and higher barriers to entry. In contrast, STOs enable companies to reach a diverse and borderless audience of potential investors, expanding their funding opportunities and driving capital from various regions worldwide.

Fractional Ownership and Investment Diversification: STOs allow fractional ownership of assets, breaking down larger assets into smaller tokenized units. This feature empowers investors to own a fraction of valuable assets that may have been otherwise out of reach. Fractional ownership lowers the entry barriers for investors, enabling them to diversify their portfolios across a broader spectrum of assets and industries.

Increased Transparency and Trust: Blockchain technology underpinning STOs introduces a high level of transparency and trust. The immutable and decentralized nature of blockchain ensures that every transaction and ownership transfer is recorded on a public ledger. This transparency builds trust between issuers and investors, reducing the risk of fraud and providing greater clarity on asset ownership and management.

Regulatory Compliance: Compared to Initial Coin Offerings (ICOs), STOs typically adhere to existing securities regulations, providing a more secure and compliant investment environment. Companies that conduct STOs must comply with the relevant securities laws, ensuring that investors' rights and protections are upheld. This regulatory compliance instills confidence in potential investors, attracting a more risk-averse and institutional investor base.

Key differences between ICO, STO, and IPO

	ICO	STO	IPO
Launching process	No need for authority approval, no compliances required. Just technical knowledge to set up a smart contract on a blockchain and a successful marketing campaign	Compliance with security laws of the relevant jurisdiction. Must be approved and regulated by national financial supervisory authorities	Companies must fulfill many compliances before launching an IPO and get approval from the appropriate authority that regulates securities and exchanges
System requirement	Quick and straightforward automated process on a blockchain	Automated process on a blockchain although regulation requires paperwork and involves bureaucracy	Lengthy process involving lawyers and underwriters
Costs	Very low	Low to medium	High
Reputation offered	Poor since 2017 ICO market due to numerous scams and rug pulls that resulted from ICOs	Very good	Very good
Rights offered	Ability to buy and sell the company's tokens to make a profit	Investors are entitled to company dividends, profit participation, voting rights, and equity	Investors get voting rights and ownership stake in the future earnings of the company
Company size	Small and medium	Small and medium	Large
Exposure	Global	While STO compliance needs a specific jurisdiction to issue tokens, investors can be global residents (abiding by the local laws adequately)	Only local within a specific jurisdiction where the IPO is generated

IEO: What is it?

An IEO or initial exchange offering refers to an event where the sale of tokens is conducted through an established mode of exchanging crypto. It is a way start-up organisations raise capital by selling utility tokens that confer preferred status with the organisation via a crypto exchange platform.

IEOs play a crucial role in the crypto ecosystem by providing investors with a more secure and regulated environment. Unlike traditional Initial Coin Offerings (ICOs), IEOs involve a cryptocurrency exchange as an intermediary, which helps to vet the projects and reduce the risk of scams.

This added layer of security attracts more investors, boosting the liquidity and visibility of new crypto projects

How Does an IEO Work?

The process of conducting an IEO on platforms like KuCoin Spotlight involves several key steps:

1. **Project Submission and Review:** The project team submits their proposal to the exchange, which includes a detailed business model, a viable use case for the technology, a robust team background, and a comprehensive whitepaper.
2. **Vetting by the Exchange:** The exchange reviews the submission thoroughly, assessing the project's potential and ensuring it meets all regulatory and security standards.
3. **Setting Fundraising Goals:** The project must decide on the structure of the IEO, including setting hard or soft caps for the fundraising.
4. **Launching the IEO:** Once approved, the exchange lists the IEO, and investors can buy tokens directly through their exchange accounts, which simplifies the process and enhances security.
5. **Post-IEO Token Listing:** After the IEO, tokens are typically listed on the exchange for trading, providing immediate liquidity for investors.

Advantages of IEO

- IEO platforms provide investors with an improved user experience in comparison to ICOs.
- There is a lesser risk of fraud, as IEOs are regulated per the exchange's set rules.
- As IEOs are conducted via third-party exchanges, they run smoothly. Exchange platforms oversee the transactions and can also provide marketing and development support.
- IEOs are not open to the public but only to the exchange platform's existing members.
- Investors are more confident when token sales occur via a trusted crypto exchange platform. This also offers the project more credibility and legitimacy, attracting further investments.

Disadvantages of IEO

- There is a risk of the crypto pump and dump scams with IEO. Scammers carry out these scams by generating a fake hype around crypto to develop interest and drive the prices so they can sell all shares soon afterwards.
- All crypto exchange platforms are not equal. Some may not be as strict in implementing regulations or doing due diligence.

Equity Token Offering (ETO):

- An equity token offering (ETO) is an event by which companies offer equity in their stock for sale to investors in the form of equity tokens.
- Equity tokens are a form of security token and store contractual information as blocks of encrypted data on their corresponding blockchain.
- During an ETO, investors buy equity tokens from the issuing company, allowing the company to raise liquid capital in order to develop its business.
- Because equity tokens are a blockchain-based equivalent of conventional equity certificates, equity token offerings have many similarities to initial public offerings (IPOs).
- Companies looking to sell equity via ETOs may subject themselves to due diligence by a third-party regulatory authority or investment bank in order to prove that they are a sound investment.
- As with IPOs, a company wishing to go public via an ETO publishes an ETO prospectus ahead of the ETO launch. This prospectus outlines the terms and conditions of attached to the ETO and to the equity tokens.

Features of Equity Tokens

1. **Programmable Equity:** The electronic value tokens allow you to grab hold of illiquid resource into a value issuance through program code.
2. **Administrative Compliance:** The blockchain application processing plant will make SEC consistent tokens and will help you in helping with respect to security runs in your purview.
3. **Liquidity:** Dilutable value tokens are expanding the liquidity for the financial specialists. Nonetheless, you can settle a secure time of 1 year from the day of Equity Token Offering.
4. **Secure Wallet:** The wallet is made as a multi-signature, multi-gadget wallet on a blockchain with egg-molded cryptography, which makes the capacity of assets safe.
5. **Worldwide Capital Investment:** The value tokens guarantee trust and dependability for institutional financial specialists to connect with the crypto-showcase, prompting a bigger pool of capital speculation.

Advantages of Equity Tokens

- **Robotized Dividends**-The ventures of the client will be naturally exchanged to the financial specialist's wallet. The wallet is completely anchored on the blockchain with exceedingly encoded coding.
- **Casting a ballot Option**-Like as the investors, the Equity token holders can likewise cast a ballot the organization highlights.
- **Benefit share rights**-The office to impart the benefits of the organization to the token holders as tokens and it tends to be straightforwardly saved into the speculator's wallet.
- **Buyback Rights**-The Company Can purchases the offers from the speculators at the market cost and offer consent to the financial specialists moreover.

Decentralized autonomous initial coin offerings (DAICOs):

- A decentralized autonomous initial coin offering (DAICO) brings the principles of [decentralized governance](#) into the fundraising campaigns of new crypto projects. The idea was proposed by [Vitalik Buterin](#), the creator of [Ethereum \(ETH\)](#), in 2018.
- The goal of DAICOs is to increase the investors' trust in [initial coin offerings \(ICOs\)](#) by allowing them to have more governance over the project funds.
- The concept of DAICO introduces the concept of the decentralized autonomous organization (DAO) smart contracts in ICOs.
- In DAICOs, all proceedings are kept locked in the smart project while the investors' group takes responsibility for the shared DAO governance.
- It is expected that DAICO will bring more democracy into the ICO campaigns and grant investors more trust in new crypto projects.
- DAICO can also act as a protective measure for investors by reducing the number of scam campaigns.

Here's an overview of how DAICOs work:

Smart Contract Controls: In a DAICO, the project's fundraising is governed by a smart contract deployed on the Ethereum blockchain. This smart contract includes features for token sale participation, fund management, and project development funding.

Token Sale Phase: Similar to ICOs, DAICOs start with a token sale phase where investors can contribute funds in exchange for project tokens. However, unlike traditional ICOs where funds are immediately released to the project team, DAICOs introduce additional controls and mechanisms.

Contribution Mechanisms: DAICOs typically implement a token sale mechanism that allows investors to contribute funds gradually over time or in predefined intervals. This helps mitigate the risk of large contributors manipulating token prices or exerting undue influence over the project.

Development Funding: After the token sale phase, DAICOs introduce a development funding mechanism where project funds are held in a smart contract-controlled treasury. Project developers must propose funding requests to the token holders, who can vote to approve or reject the release of funds for specific project milestones or development tasks.

Refund Mechanism: In the event that token holders are dissatisfied with the project's progress or performance, DAICOs may include a refund mechanism that allows investors to vote for a refund of remaining funds held in the treasury. This provides an additional layer of investor protection and incentivizes project teams to meet their development milestones.

ERC-721:

ERC or Ethereum Request for Comments is simply a guideline or a template that all Ethereum-based tokens should follow. [Ethereum's smart contract](#) programmers are accountable for writing ERC-related documents to describe the set of rules that every Ethereum-based Token must follow.

A token is basically a digital asset that represents something. It can be traded, sold, or bought by a user. It is a kind of virtual currency. They are used for investment purposes, to store value, to make purchases, etc. These are used to facilitate transactions along a blockchain.

ERC-721 is basically a template or a guideline that other developers agree to follow. It is a widely used standard. Being a widely used standard also means being compatible with a wide range of applications.

ERC-721 is a standard for “non-fungible” tokens. A non-fungible token is the type of token that is unique. These are cryptographic assets on a blockchain that has a unique code and metadata differentiating them from one another.

The ERC-721 standard non-fungible tokens can be seen in blockchain-based games where each asset is unique and players can trade it and sell or buy items with it.

One of the examples that use the ERC-721 non-fungible tokens standard is the Cryptokitties game which is basically a game developed on the blockchain which allows users to sell, buy and breed virtual cats.

Some other examples include Ethermon, MyCrypto, Cryptodoggies, etc. The assets bought with a non-fungible token are unique and only belong to the owner. It can't be traded. It's like a property that one owns (a house).

The ERC-721 standard tokens are created the same way as other kinds of tokens and can be created as many as the user wishes but the only difference is that the tokens have a unique value and unique metadata, making it non-fungible.

Characteristics of ERC-721 Tokens:

- ERC-721 tokens are the standards for non-fungible tokens (NFTs).
- These tokens cannot be exchanged for anything of equal value since they are unique in themselves, representing a unique asset.
- Each NFT token is linked to different owners and has its own tokenID or metadata that makes them unique.
- The most popular application areas of ERC-721 tokens are NFTs in gaming (such as cryptokitties).

Who Invented ERC-721?

- The initial ERC-721 specification was proposed by Dieter Shirley as an Ethereum Improvement Proposal (EIP), which is a process for introducing new standards to Ethereum, in January 2018.
- ERC-721 was Ethereum's first NFT token standard.
- It laid the groundwork for the uses of NFTs we see today

ERC 1155:

- Introduced in 2018, ERC-1155 is a multi-token standard that provides a standard interface for smart contracts managing multiple types of tokens. The original intention behind the creation of ERC-1155 involved addressing the challenges experienced by developers in the world of blockchain-based gaming.
- ERC-1155 improves upon the functionality of ERC-721, increasing its efficiency and correcting evident implementation errors. It can support an infinite quantity of tokens in a single contract, underpinning fungible, semi-fungible, and NFTs, which can transition between both of the former during its lifecycle.
- ERC-1155 also has a security function that verifies a transaction's validity, allowing tokens to be returned to the issuer if a transaction fails. This eliminates the anxiety of potential user mistakes such as sending tokens to the wrong address. This function, along with other rules, safeguards against exploitation.

Feature	ERC-721	ERC-1155
Token Type	Non-fungible token (NFT)	Multi-fungible token (MFT)
Standard	ERC-721	ERC-1155
Ownership	Each token is unique and indivisible	Tokens can be fungible (identical) or unique
Use Case	Typically used for unique digital assets	Versatile; supports both unique and fungible
Storage	Requires separate contract for each token	Efficient; multiple tokens stored in one contract
Transfers	Individual token transfers	Batch transfers for multiple tokens
Gas Costs	Higher gas costs due to individual transfers	Lower gas costs for batch transfers
Metadata	Individual metadata for each token	Shared metadata for multiple tokens
Interoperability	Limited interoperability between platforms	Enhanced interoperability across platforms